side. It is easy to deduce from Newton's laws that F_v is linear in v, so there exists a linear operator $S_P : V \to V$ such that $F_v = S_P v$. It is called **the stress tensor**.

An elasticity law is an equation $S_P = f(d_P)$, where f is a function. The simplest such law is a linear law (Hooke's law): $f : S^2V \to \mathrm{End}(V)$ is a linear function. In general, such a function is defined by $9 \cdot 6 = 54$ parameters, but we will show there are actually only two essential ones — the **compression modulus** K and the **shearing modulus** μ. For this purpose we will use representation theory.

Recall that the group $SO(3)$ of rotations acts on V, so S^2V, $\mathrm{End}(V)$ are representations of this group. The laws of physics must be invariant under this group (Galileo transformations), so f must be a homomorphism of representations.

(a) Show that $\mathrm{End}(V)$ admits a decomposition $\mathbb{R} \oplus V \oplus W$, where $\mathbb{R}$ is the trivial representation, V is the standard 3-dimensional representation, and W is a 5-dimensional representation of SO(3). Show that $S^2V = \mathbb{R} \oplus W$.

(b) Show that V and W are irreducible, even after complexification. Deduce using Schur's lemma that S_P is always symmetric, and for $x \in \mathbb{R}, y \in W$ one has $f(x + y) = Kx + \mu y$ for some real numbers K, μ.

In fact, it is clear from physics that K, μ are positive. Physically, the compression modulus K characterizes resistance of the material to compression or dilation, while the shearing modulus μ characterizes its resistance to changing the shape of the object without changing its volume. For instance, clay (used for sculpting) has a large compression modulus but a small shearing modulus.

4.13. Historical interlude: William Rowan Hamilton's quaternion of geometry, algebra, metaphysics, and poetry

At age 17, William Rowan Hamilton's interest in mathematics was sparked by his discovery of an error in Laplace's *Celestial Mechanics*. The Royal Astronomer of Ireland was so impressed that in a few years he secured for Hamilton the appointment as Royal Astronomer